CryptoRadar

Multi-Dataset Cryptocurrency Risk Intelligence Platform

Cybercrime Trend Analysis Using Machine Learning: Cryptocurrency Scams and Rug-Pull Fraud

Academic research prototype. Generated from the training run of 2026-10-15T04:27:10+00:00 (UTC).

Abstract

This report documents an offline research prototype that builds separate machine-learning models for four cryptocurrency datasets: SolRPDS (Solana liquidity pools), Real-CATS Bitcoin addresses, Real-CATS Ethereum addresses and the Elliptic Bitcoin transaction graph. 7 models were trained in the run that produced this report. All splits are time-aware, preprocessing is fitted on training data only, and SHAP is used to explain the main model of each dataset. Every number below is read from files saved by the training run. No cross-dataset generalisation is claimed.

The positive class is an inactivity-based proxy derived from SolRPDS, not an independently verified rug-pull ground truth.

1. Scope and Terminology

Model outputs are called model probability, model risk score and model-indicated risk. They are uncalibrated and are not confirmations of fraud. Scores are uncalibrated model probabilities: no calibration map was fitted, so a score of 0.80 does not mean an 80% chance of anything. They are useful for ranking within one model and are not comparable across datasets.

2. Datasets and Data Audit

2.1 SolRPDS

The yearly SolRPDS files are partitions of one dataset and were loaded and concatenated. The target is the inactivity-based rug-pull proxy label: y = (INACTIVITY_STATUS == 'Inactive').

The positive class is an inactivity-based proxy derived from SolRPDS, not an independently verified rug-pull ground truth.

Item	Value
Files loaded	2021.json, 2022.json, 2023.json, Jan_2024-Nov_2024.json
Files missing	none
Rows (after de-duplication)	116,304
Exact duplicate rows dropped	3
Unique pools	63,520
Proxy-positive rows (Inactive)	22,555
Proxy-negative rows (Active)	93,749

2.2 Real-CATS BTC

Unit assumption: satoshi -> BTC (/1e8). Implied USD per coin from the dataset's own USD columns (median): 35813.60, used as a sanity check of that assumption.

Class	Rows	Zero-transaction rows	Zero-transaction share (%)	Undated rows
criminal	90,597	50,565	55.81	50,565
benign	90,164	3,581	3.97	3,581

A model trained on all addresses may learn that no observed activity means criminal, which is not a behavioural fraud signal.

2.3 Real-CATS ETH

Unit assumption: Wei -> ETH (/1e18). Implied USD per coin from the dataset's own USD columns (median): 1883.11, used as a sanity check of that assumption.

Class	Rows	Zero-transaction rows	Zero-transaction share (%)	Undated rows
criminal	12,572	1,661	13.21	1,661
benign	16,019	424	2.65	424

2.4 Elliptic

203,769 transactions, 234,355 edges, time steps 1-49. Class counts: illicit 4,545, licit 42,019, unknown 157,205. Unknown transactions are not treated as licit and are excluded from training, validation and testing.

Split	Time steps	Labelled	Illicit	Licit
train	1-29	26,381	2,871	23,510
val	30-34	3,513	591	2,922
test	35-49	16,670	1,083	15,587

3. Methodology

3.1 Separate models

Each dataset has its own model. Datasets are never merged, scores from different models are not comparable, and no cross-dataset generalisation is claimed.

3.2 Splits

SolRPDS and Real-CATS use chronological splits on first activity time with fractions 60% / 20% / 20% (train / validation / test). SolRPDS rows are grouped by pool address so a pool cannot appear in two splits. Elliptic uses time steps 1-29 for training, 30-34 for validation and 35-49 for testing, labelled transactions only.

3.3 Leakage controls

Preprocessing (median imputation, and log scaling plus standardisation for the logistic baseline) is fitted on the training split only. The decision threshold is selected on the validation split only, and the test split is scored once. Identifiers, label subtypes, collection or verification flags (watchback_checked, gs_checked, etherscan_checked), all *_transaction_id columns and absolute timestamps are never model features. SolRPDS LAST_SWAP_TIMESTAMP, LAST_SWAP_TX_ID and everything derived from them are excluded from the primary model.

3.4 Models and metrics

Each dataset is modelled with logistic regression and random forest baselines and a gradient-boosted tree main model (XGBoost) with class weighting for imbalance. Hyper-parameters are fixed defaults. Headline metrics for the positive class are PR-AUC (average precision), F1, precision, recall, ROC-AUC and MCC. Accuracy is not reported as a headline metric because the classes are imbalanced.

Scores are uncalibrated model probabilities: no calibration map was fitted, so a score of 0.80 does not mean an 80% chance of anything. They are useful for ranking within one model and are not comparable across datasets.

4. Results

Results are reported per dataset and per model variant. Variants of the same dataset (primary vs ablation, activity-filtered vs full) are separate results and are never averaged or merged.

SolRPDS - Primary model - Excludes outcome-related timestamp fields

The positive class is an inactivity-based proxy derived from SolRPDS, not an independently verified rug-pull ground truth.

The main model (xgboost) uses a decision threshold of 0.5369 (maximises validation F1). On the test split (23,261 rows, 2,492 positives) it obtained PR-AUC 0.444, F1 0.418, precision 0.301 and recall 0.685. On the validation split (n=23,260) PR-AUC was 0.681. Probabilities are uncalibrated.

Split	Rows	Positives	Negatives	Positive rate
train	69,783	14,857	54,926	0.213

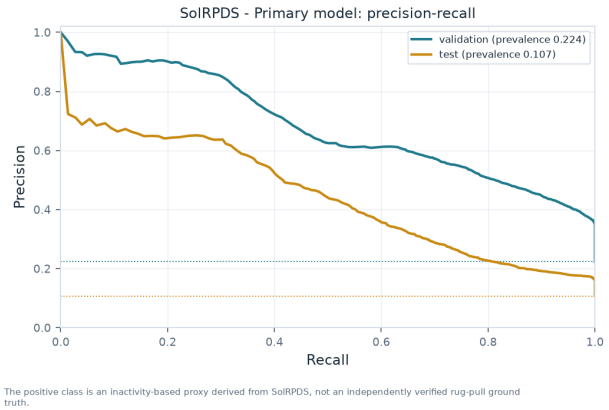
Split	Rows	Positives	Negatives	Positive rate
val	23,260	5,206	18,054	0.224
test	23,261	2,492	20,769	0.107

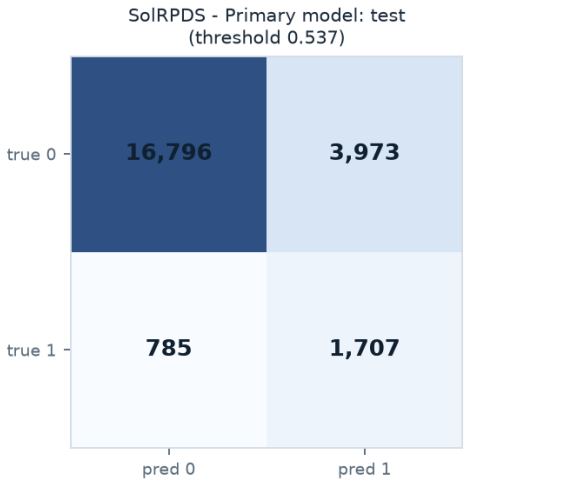
Metric (positive class)	Train (in-sample)	Validation	Test	Test minus validation
PR-AUC (average precision)	0.741	0.681	0.444	-0.237
F1 (positive class)	0.685	0.632	0.418	-0.214
Precision	0.574	0.543	0.301	-0.242
Recall	0.849	0.757	0.685	-0.072
ROC-AUC	0.922	0.881	0.844	-0.037
MCC	0.598	0.515	0.355	-0.160

Calibration diagnostics on the test split (the model is not recalibrated): Brier score 0.162, expected calibration error (10 bins) 0.235.

Test-split metrics of every algorithm fitted on identical splits:

Algorithm	PR-AUC	F1	Precision	Recall	ROC-AUC	MCC
logistic_regression	0.435	0.310	0.186	0.929	0.848	0.274
random_forest	0.396	0.354	0.230	0.768	0.828	0.297
xgboost (main)	0.444	0.418	0.301	0.685	0.844	0.355





SolRPDS - Ablation model - Includes timestamp-derived duration features; potentially label-leaking

The positive class is an inactivity-based proxy derived from SolRPDS, not an independently verified rug-pull ground truth.

This ablation model includes timestamp-derived duration features and is potentially label-leaking. It is reported to quantify how much of the result depends on those fields and must not be read as a fair performance estimate.

The main model (xgboost) uses a decision threshold of 0.6200 (maximises validation F1). On the test split (23,261 rows, 2,492 positives) it obtained PR-AUC 0.421, F1 0.484, precision 0.370 and recall 0.701. On the validation split (n=23,260) PR-AUC was 0.581. Probabilities are uncalibrated.

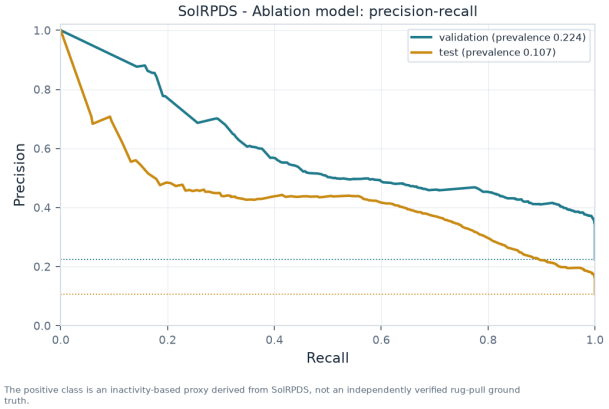
Split	Rows	Positives	Negatives	Positive rate
train	69,783	14,857	54,926	0.213
val	23,260	5,206	18,054	0.224
test	23,261	2,492	20,769	0.107

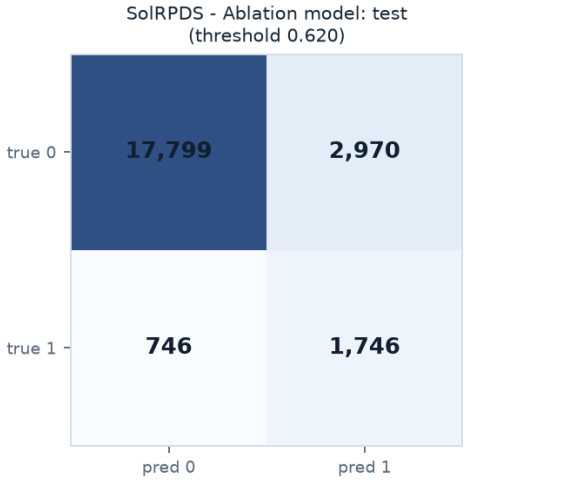
Metric (positive class)	Train (in-sample)	Validation	Test	Test minus validation
PR-AUC (average precision)	0.795	0.581	0.421	-0.161
F1 (positive class)	0.711	0.584	0.484	-0.100
Precision	0.760	0.468	0.370	-0.098
Recall	0.669	0.778	0.701	-0.077
ROC-AUC	0.934	0.841	0.867	0.026
MCC	0.641	0.451	0.429	-0.022

Calibration diagnostics on the test split (the model is not recalibrated): Brier score 0.210, expected calibration error (10 bins) 0.356.

Test-split metrics of every algorithm fitted on identical splits:

Algorithm	PR-AUC	F1	Precision	Recall	ROC-AUC	MCC
logistic_regression	0.492	0.385	0.250	0.842	0.869	0.347
random_forest	0.355	0.443	0.344	0.620	0.857	0.375
xgboost (main)	0.421	0.484	0.370	0.701	0.867	0.429





Real-CATS BTC - Activity-filtered - Headline BTC result; excludes zero-transaction addresses

Headline BTC result. Metrics here are reported separately from the Full BTC model and are never averaged or combined with them.

The main model (xgboost) uses a decision threshold of 0.7979 (maximises validation F1). On the test split (25,310 rows, 920 positives) it obtained PR-AUC 0.223, F1 0.317, precision 0.265 and recall 0.393. On the validation split (n=25,336) PR-AUC was 0.468. Probabilities are uncalibrated.

Split	Rows	Positives	Negatives	Positive rate
train	75,969	37,194	38,775	0.490
val	25,336	1,918	23,418	0.076
test	25,310	920	24,390	0.036

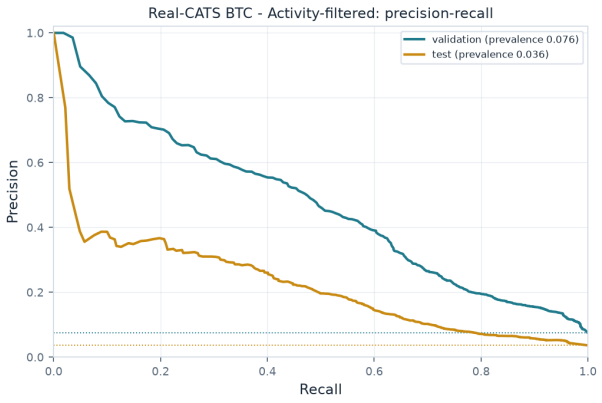
Metric (positive class)	Train (in-sample)	Validation	Test	Test minus validation
PR-AUC (average precision)	0.929	0.468	0.223	-0.245
F1 (positive class)	0.680	0.487	0.317	-0.170
Precision	0.952	0.521	0.265	-0.256
Recall	0.529	0.457	0.393	-0.064

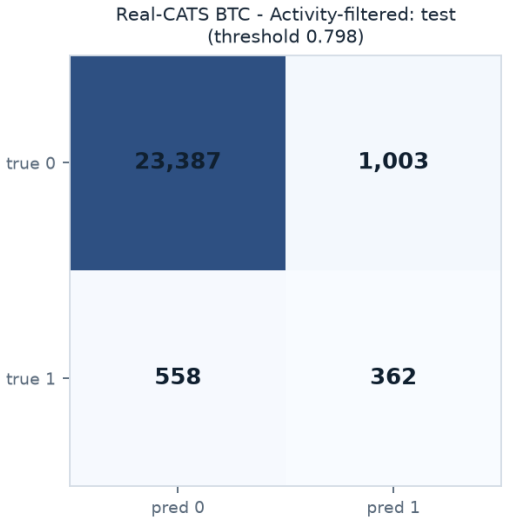
Metric (positive class)	Train (in-sample)	Validation	Test	Test minus validation
ROC-AUC	0.933	0.864	0.809	-0.055
MCC	0.566	0.449	0.292	-0.157

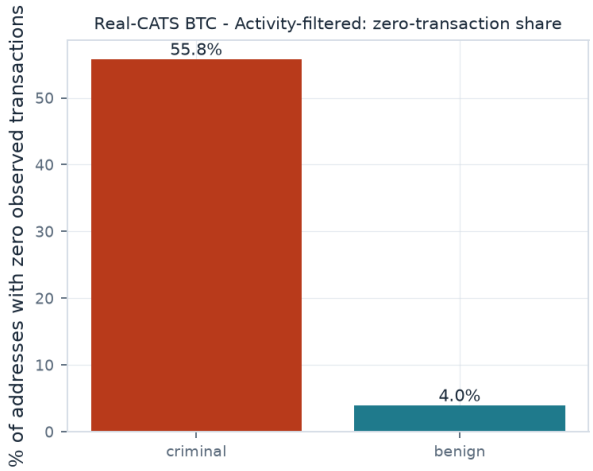
Calibration diagnostics on the test split (the model is not recalibrated): Brier score 0.212, expected calibration error (10 bins) 0.370.

Test-split metrics of every algorithm fitted on identical splits:

Algorithm	PR-AUC	F1	Precision	Recall	ROC-AUC	MCC
logistic_regression	0.111	0.176	0.117	0.360	0.694	0.152
random_forest	0.143	0.157	0.089	0.648	0.769	0.169
xgboost (main)	0.223	0.317	0.265	0.393	0.809	0.292







Report-only: zero-activity addresses (no observed on-chain activity)

54,146 addresses have no observed transactions and no timestamps. They were never seen by this model. All their features are zero, so the model cannot tell criminal from benign among them; the shares below reflect the score assigned to the all-zero feature vector.

Class	Rows	Share above threshold	Mean model risk score
criminal	50,565	0.000	0.522
benign	3,581	0.000	0.522

Real-CATS BTC - Full - Zero-activity addresses are training-only and flagged report-only

Addresses with zero observed on-chain transactions are marked report-only / no observed on-chain activity. In the Full BTC model a classifier can learn that 'no observed activity = criminal', which is a property of how the dataset was assembled and not a behavioural fraud signal. The activity-filtered BTC model is the headline BTC result.

The main model (xgboost) uses a decision threshold of 0.6861 (maximises validation F1). On the test split (25,310 rows, 920 positives) it obtained PR-AUC 0.208, F1 0.297, precision 0.227 and recall 0.429. On the validation split (n=25,336) PR-AUC was 0.437. Probabilities are uncalibrated.

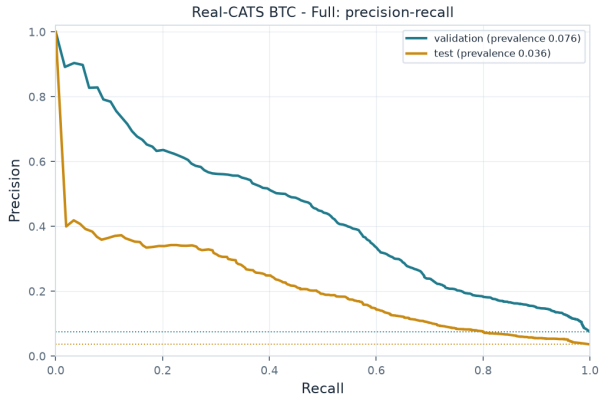
Split	Rows	Positives	Negatives	Positive rate
train	75,969	37,194	38,775	0.490
val	25,336	1,918	23,418	0.076
test	25,310	920	24,390	0.036
train (actually used, incl. zero-activity training-only rows)	130,115	87,759	42,356	0.674

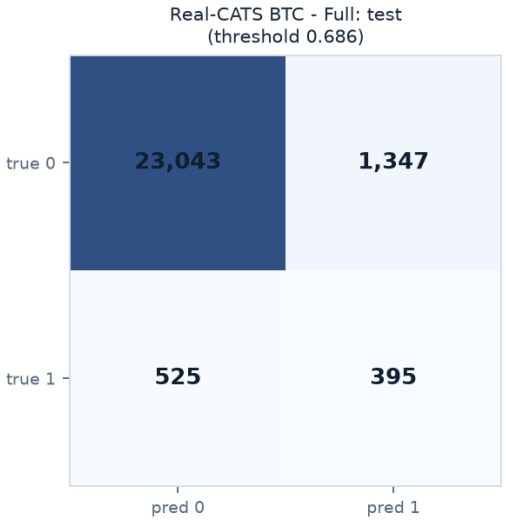
Metric (positive class)	Train (in-sample)	Validation	Test	Test minus validation
PR-AUC (average precision)	0.939	0.437	0.208	-0.229
F1 (positive class)	0.865	0.474	0.297	-0.177
Precision	0.940	0.474	0.227	-0.247
Recall	0.801	0.474	0.429	-0.045
ROC-AUC	0.921	0.855	0.810	-0.045
MCC	0.659	0.431	0.277	-0.155

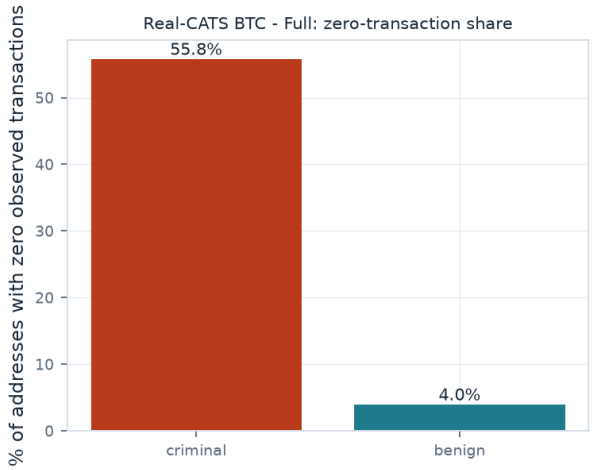
Calibration diagnostics on the test split (the model is not recalibrated): Brier score 0.132, expected calibration error (10 bins) 0.261.

Test-split metrics of every algorithm fitted on identical splits:

Algorithm	PR-AUC	F1	Precision	Recall	ROC-AUC	MCC
logistic_regression	0.114	0.165	0.106	0.373	0.666	0.143
random_forest	0.138	0.174	0.102	0.602	0.771	0.183
xgboost (main)	0.208	0.297	0.227	0.429	0.810	0.277







Report-only: zero-activity addresses (no observed on-chain activity)

54,146 addresses have no observed transactions and no timestamps. They were training-only rows in this model, so these scores are in-sample diagnostics. All their features are zero, so the model cannot tell criminal from benign among them; the shares below reflect the score assigned to the all-zero feature vector.

Class	Rows	Share above threshold	Mean model risk score
criminal	50,565	1.000	0.864
benign	3,581	1.000	0.864

Real-CATS ETH - Addresses with observed on-chain activity

The main model (xgboost) uses a decision threshold of 0.3000 (maximises validation F1). On the test split (5,301 rows, 1,060 positives) it obtained PR-AUC 0.702, F1 0.634, precision 0.539 and recall 0.768. On the validation split (n=5,301) PR-AUC was 0.863. Probabilities are uncalibrated.

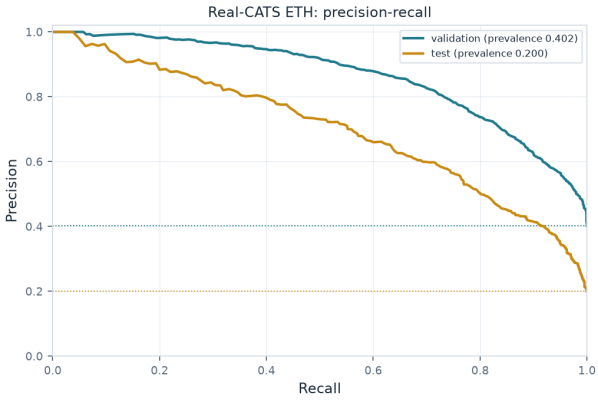
Split	Rows	Positives	Negatives	Positive rate
train	15,904	7,722	8,182	0.486
val	5,301	2,129	3,172	0.402
test	5,301	1,060	4,241	0.200

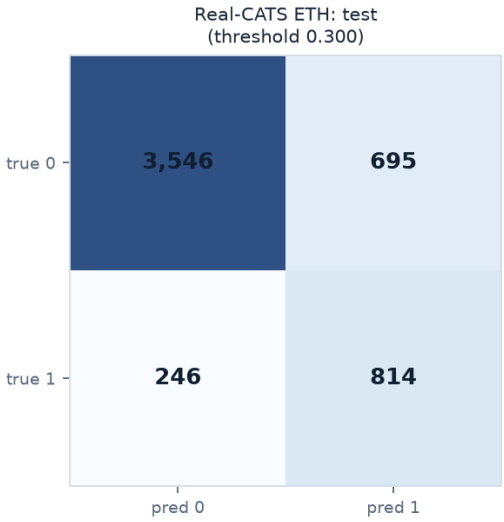
Metric (positive class)	Train (in-sample)	Validation	Test	Test minus validation
PR-AUC (average precision)	1.000	0.863	0.702	-0.161
F1 (positive class)	0.990	0.771	0.634	-0.137
Precision	0.984	0.724	0.539	-0.185
Recall	0.995	0.824	0.768	-0.056
ROC-AUC	1.000	0.891	0.888	-0.004
MCC	0.980	0.604	0.535	-0.068

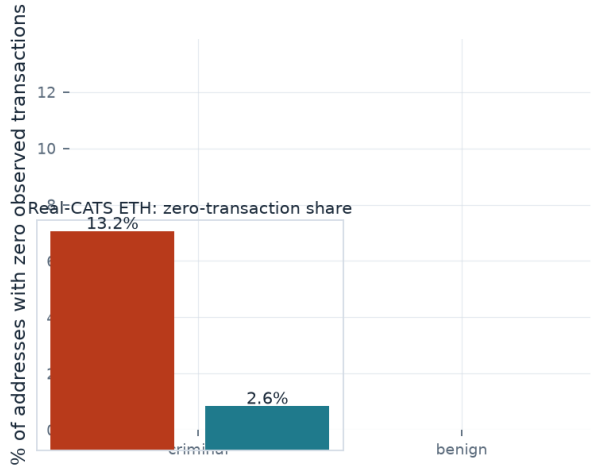
Calibration diagnostics on the test split (the model is not recalibrated): Brier score 0.108, expected calibration error (10 bins) 0.059.

Test-split metrics of every algorithm fitted on identical splits:

Algorithm	PR-AUC	F1	Precision	Recall	ROC-AUC	MCC
logistic_regression	0.606	0.518	0.373	0.850	0.834	0.396
random_forest	0.608	0.546	0.439	0.721	0.830	0.418
xgboost (main)	0.702	0.634	0.539	0.768	0.888	0.535







Report-only: zero-activity addresses (no observed on-chain activity)

2,085 addresses have no observed transactions and no timestamps. They were never seen by this model. All their features are zero, so the model cannot tell criminal from benign among them; the shares below reflect the score assigned to the all-zero feature vector.

Class	Rows	Share above threshold	Mean model risk score
criminal	1,661	1.000	0.366
benign	424	1.000	0.352

Elliptic - Labelled transactions only (unknown excluded)

The main model (xgboost) uses a decision threshold of 0.4005 (maximises validation F1). On the test split (16,670 rows, 1,083 positives) it obtained PR-AUC 0.790, F1 0.734, precision 0.731 and recall 0.737. On the validation split (n=3,513) PR-AUC was 0.989. Probabilities are uncalibrated.

Split	Rows	Positives	Negatives	Positive rate
train	26,381	2,871	23,510	0.109
val	3,513	591	2,922	0.168
test	16,670	1,083	15,587	0.065

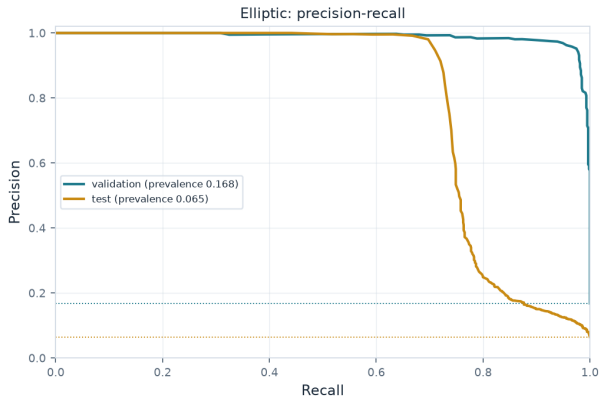
Metric (positive class)	Train (in-sample)	Validation	Test	Test minus validation
PR-AUC (average precision)	1.000	0.989	0.790	-0.199
F1 (positive class)	1.000	0.966	0.734	-0.232

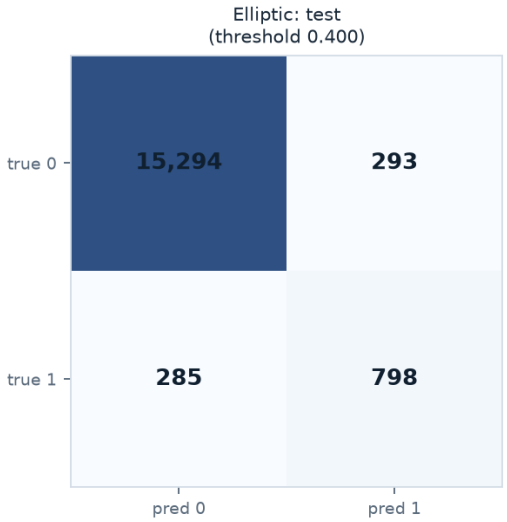
Metric (positive class)	Train (in-sample)	Validation	Test	Test minus validation
Precision	1.000	0.957	0.731	-0.225
Recall	1.000	0.975	0.737	-0.238
ROC-AUC	1.000	0.998	0.919	-0.078
MCC	1.000	0.959	0.716	-0.243

Calibration diagnostics on the test split (the model is not recalibrated): Brier score 0.026, expected calibration error (10 bins) 0.028.

Test-split metrics of every algorithm fitted on identical splits:

Algorithm	PR-AUC	F1	Precision	Recall	ROC-AUC	MCC
logistic_regression	0.246	0.367	0.243	0.751	0.867	0.362
random_forest	0.789	0.795	0.889	0.718	0.934	0.787
xgboost (main)	0.790	0.734	0.731	0.737	0.919	0.716





Elliptic - Ablation - Adds in/out-degree computed from the edge list

The main model (xgboost) uses a decision threshold of 0.4187 (maximises validation F1). On the test split (16,670 rows, 1,083 positives) it obtained PR-AUC 0.787, F1 0.743, precision 0.747 and recall 0.739. On the validation split (n=3,513) PR-AUC was 0.986. Probabilities are uncalibrated.

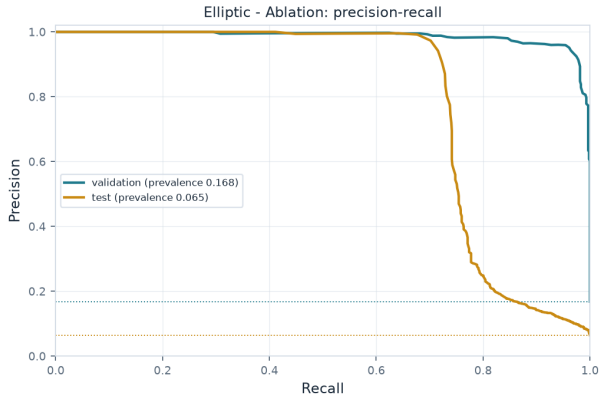
Split	Rows	Positives	Negatives	Positive rate
train	26,381	2,871	23,510	0.109
val	3,513	591	2,922	0.168
test	16,670	1,083	15,587	0.065

Metric (positive class)	Train (in-sample)	Validation	Test	Test minus validation
PR-AUC (average precision)	1.000	0.986	0.787	-0.199
F1 (positive class)	1.000	0.959	0.743	-0.216
Precision	1.000	0.958	0.747	-0.211
Recall	1.000	0.959	0.739	-0.221
ROC-AUC	1.000	0.997	0.913	-0.085
MCC	1.000	0.950	0.725	-0.225

Calibration diagnostics on the test split (the model is not recalibrated): Brier score 0.026, expected calibration error (10 bins) 0.028.

Test-split metrics of every algorithm fitted on identical splits:

Algorithm	PR-AUC	F1	Precision	Recall	ROC-AUC	MCC
logistic_regression	0.251	0.360	0.236	0.765	0.870	0.358
random_forest	0.786	0.737	0.741	0.733	0.932	0.719
xgboost (main)	0.787	0.743	0.747	0.739	0.913	0.725



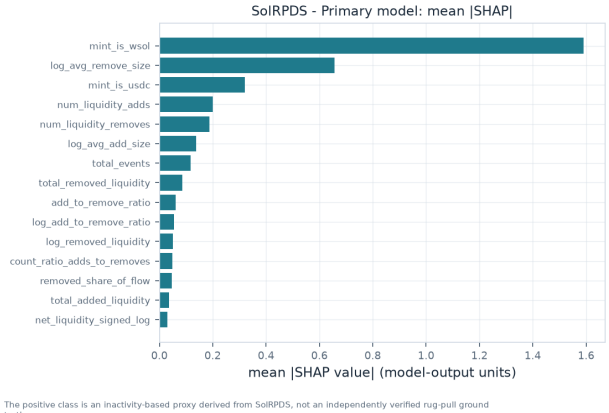


5. Explainability (SHAP)

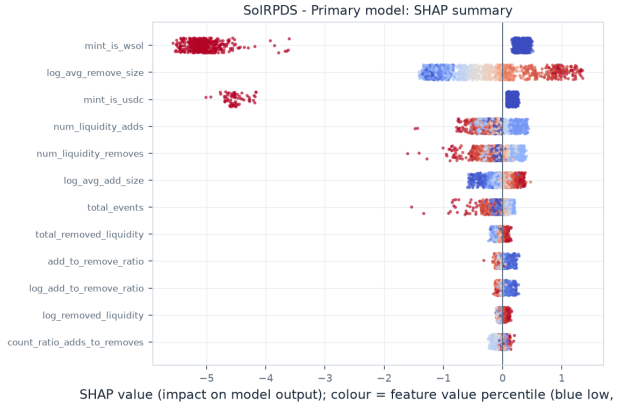
SHAP values describe how the trained model's output depends on its inputs. They are associative explanations of the model, not causal claims about why an address, transaction or pool is risky.

SolRPDS - Primary model - Excludes outcome-related timestamp fields

Rank	Feature	Mean |SHAP|
1	mint_is_wsol	1.59088
2	log_avg_remove_size	0.65742
3	mint_is_usdc	0.32137
4	num_liquidity_adds	0.19973
5	num_liquidity_removes	0.18842
6	log_avg_add_size	0.13772
7	total_events	0.11763
8	total_removed_liquidity	0.08497



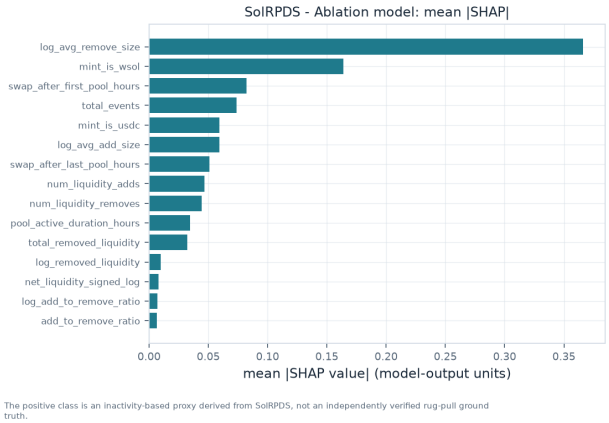
The positive class is an inactivity-based proxy derived from SolRPDS, not an independently verified rug-pull ground truth.



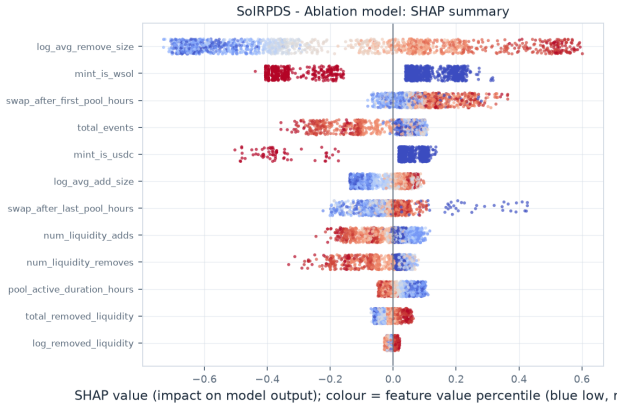
The positive class is an inactivity-based proxy derived from SolRPDS, not an independently verified rug-pull ground truth.

SolRPDS - Ablation model - Includes timestamp-derived duration features; potentially label-leaking

Rank	Feature	Mean |SHAP|
1	log_avg_remove_size	0.36624
2	mint_is_wsol	0.16391
3	swap_after_first_pool_hours	0.08227
4	total_events	0.07384
5	mint_is_usdc	0.05944
6	log_avg_add_size	0.05940
7	swap_after_last_pool_hours	0.05095
8	num_liquidity_adds	0.04679



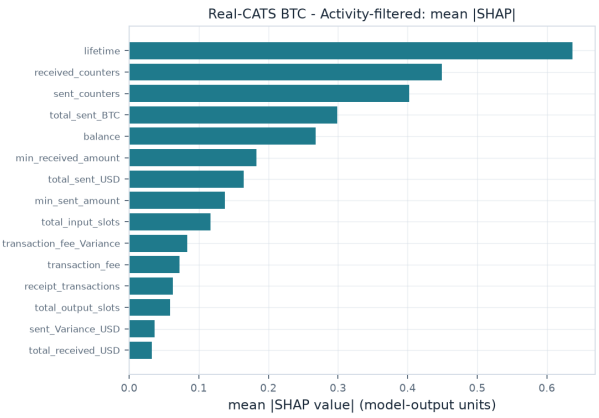
The positive class is an inactivity-based proxy derived from SolRPDS, not an independently verified rug-pull ground truth.

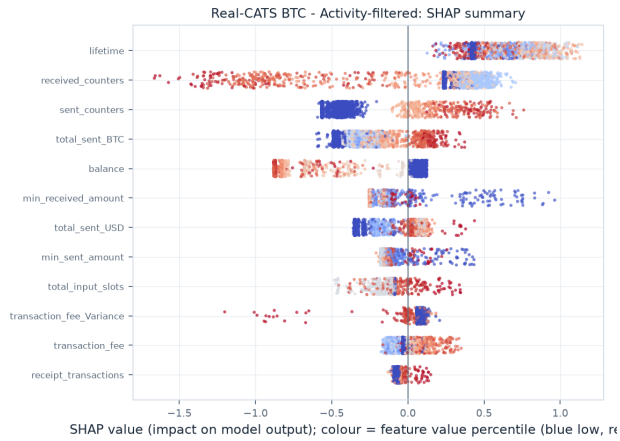


The positive class is an inactivity-based proxy derived from SolRPDS, not an independently verified rug-pull ground truth.

Real-CATS BTC - Activity-filtered - Headline BTC result; excludes zero-transaction addresses

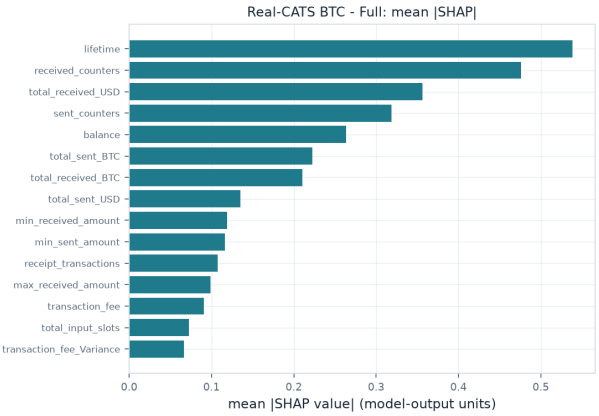
Rank	Feature	Mean |SHAP|
1	lifetime	0.63720
2	received_counters	0.44926
3	sent_counters	0.40250
4	total_sent_BTC	0.29870
5	balance	0.26761
6	min_received_amount	0.18292
7	total_sent_USD	0.16434
8	min_sent_amount	0.13792

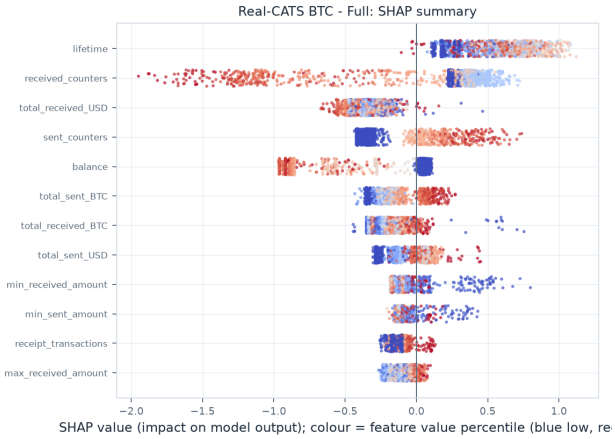




Real-CATS BTC - Full - Zero-activity addresses are training-only and flagged report-only

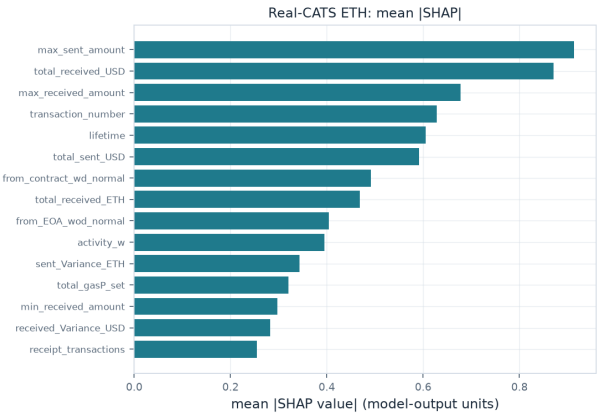
Rank	Feature	Mean |SHAP|
1	lifetime	0.53869
2	received_counters	0.47614
3	total_received_USD	0.35621
4	sent_counters	0.31842
5	balance	0.26349
6	total_sent_BTC	0.22243
7	total_received_BTC	0.21035
8	total_sent_USD	0.13542

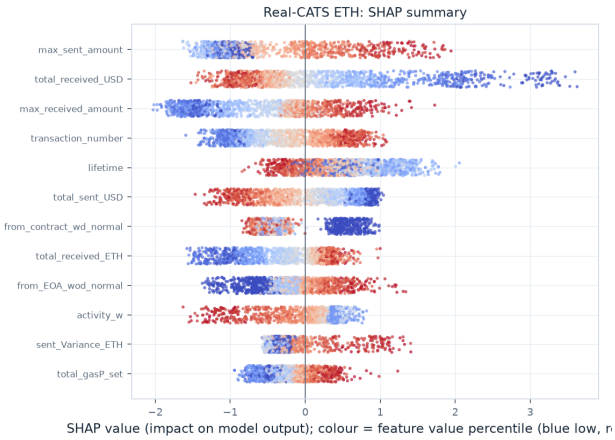




Real-CATS ETH - Addresses with observed on-chain activity

Rank	Feature	Mean |SHAP|
1	max_sent_amount	0.91303
2	total_received_USD	0.87074
3	max_received_amount	0.67745
4	transaction_number	0.62773
5	lifetime	0.60473
6	total_sent_USD	0.59181
7	from_contract_wd_normal	0.49199
8	total_received_ETH	0.46823

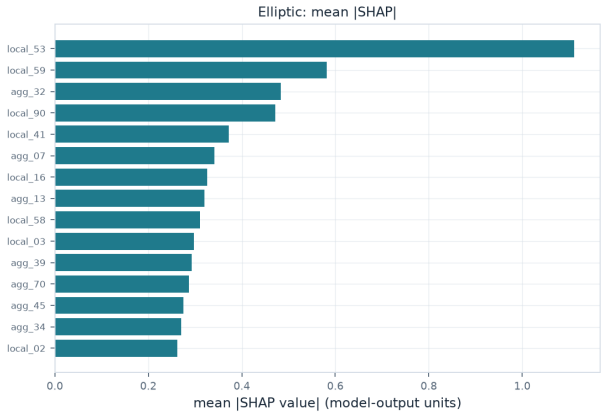


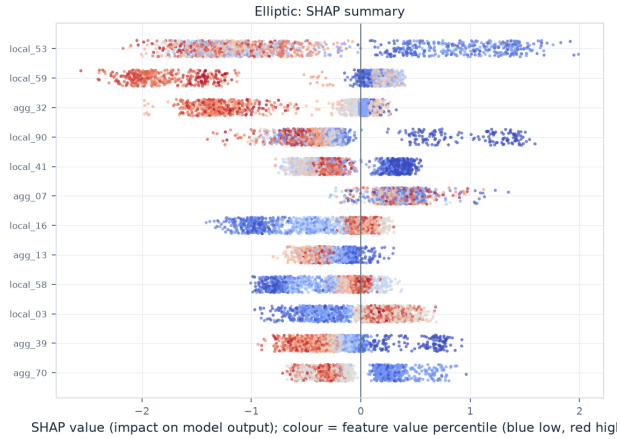


Elliptic - Labelled transactions only (unknown excluded)

Rank	Feature	Mean |SHAP|
1	local_53	1.11047

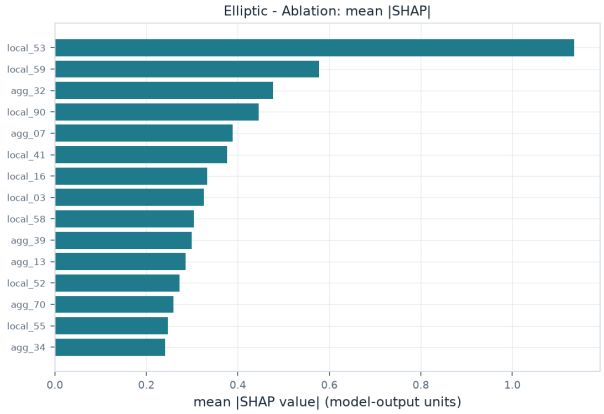
Rank	Feature	Mean |SHAP|
2	local_59	0.58145
3	agg_32	0.48378
4	local_90	0.47157
5	local_41	0.37190
6	agg_07	0.34136
7	local_16	0.32598
8	agg_13	0.32036

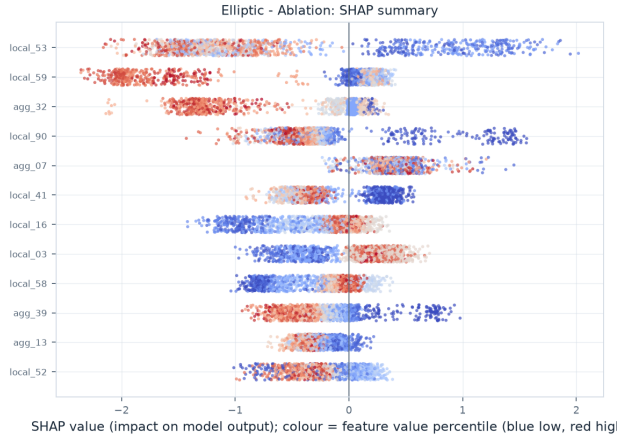




Elliptic - Ablation - Adds in/out-degree computed from the edge list

Rank	Feature	Mean |SHAP|
1	local_53	1.13457
2	local_59	0.57689
3	agg_32	0.47748
4	local_90	0.44515
5	agg_07	0.38886
6	local_41	0.37688
7	local_16	0.33289
8	local_03	0.32546

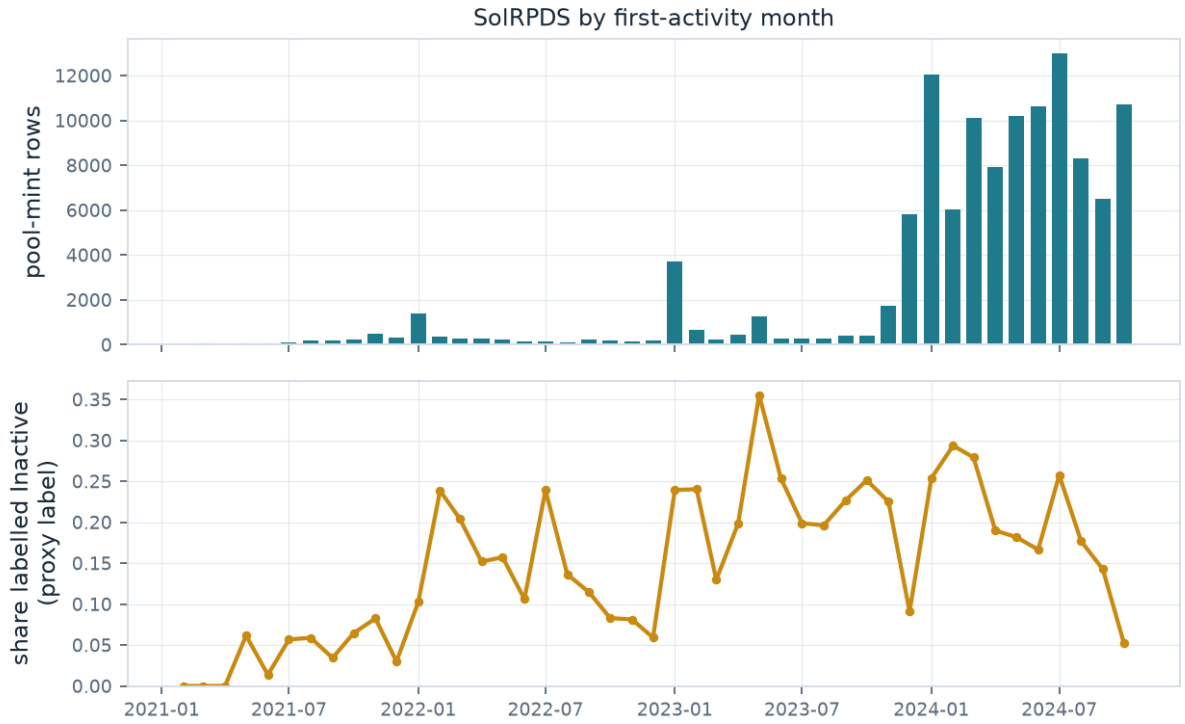




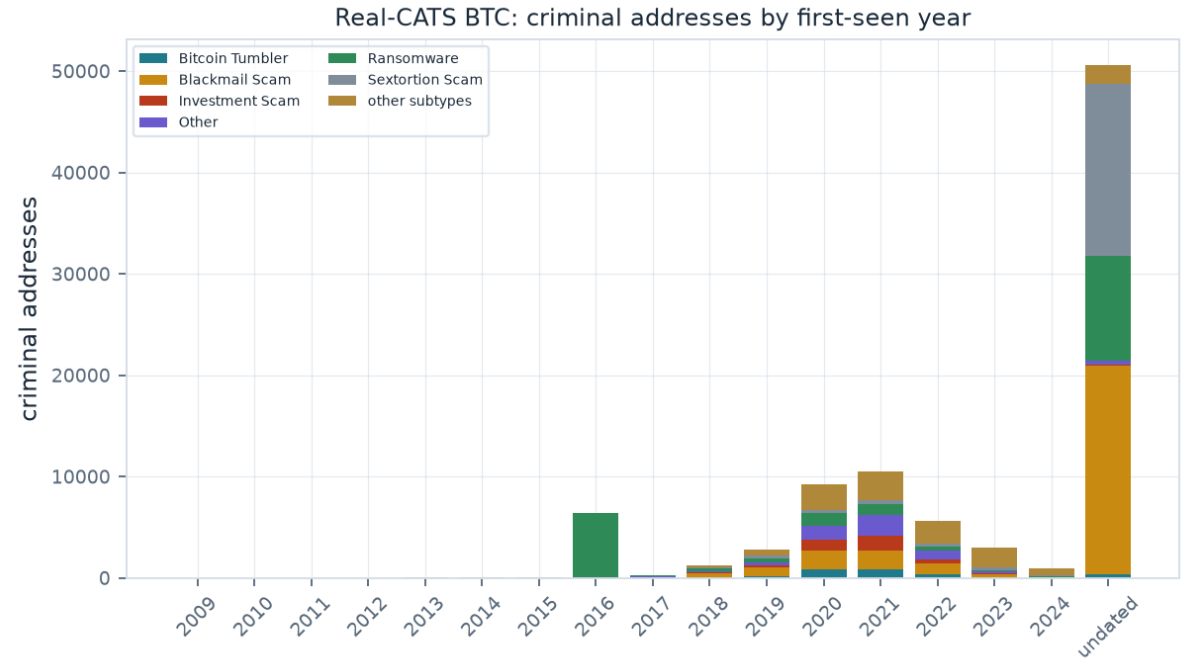
6. Trend Analysis

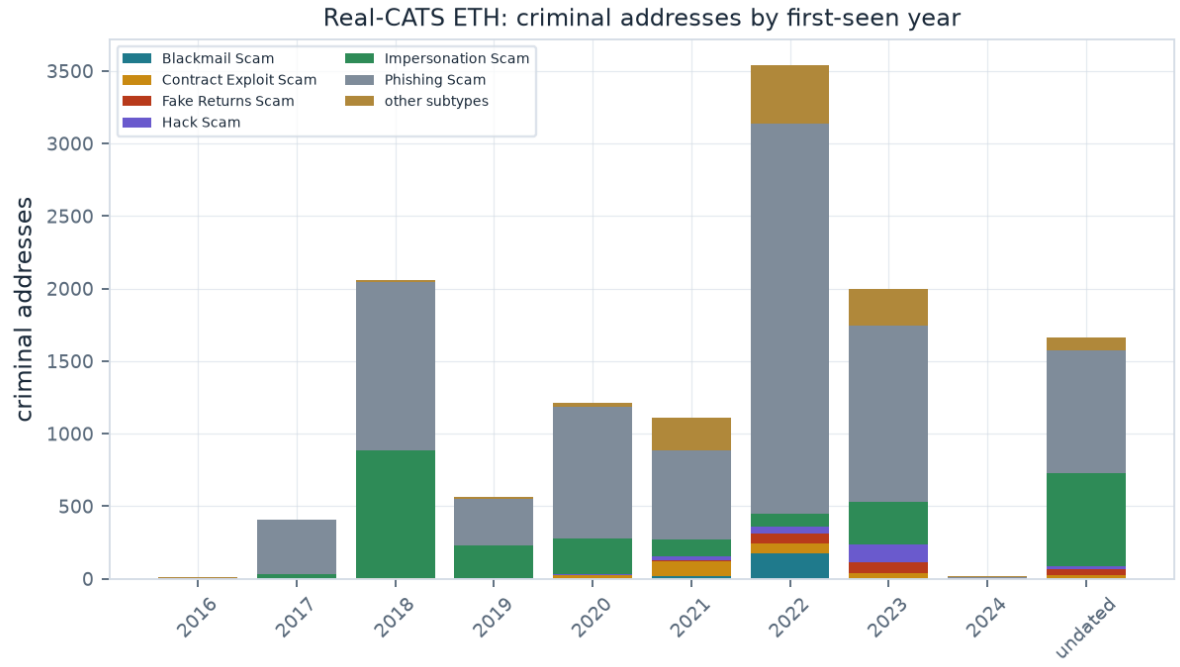
Trends are descriptive counts and shares by period; no model is involved. Datasets are shown separately and never merged.

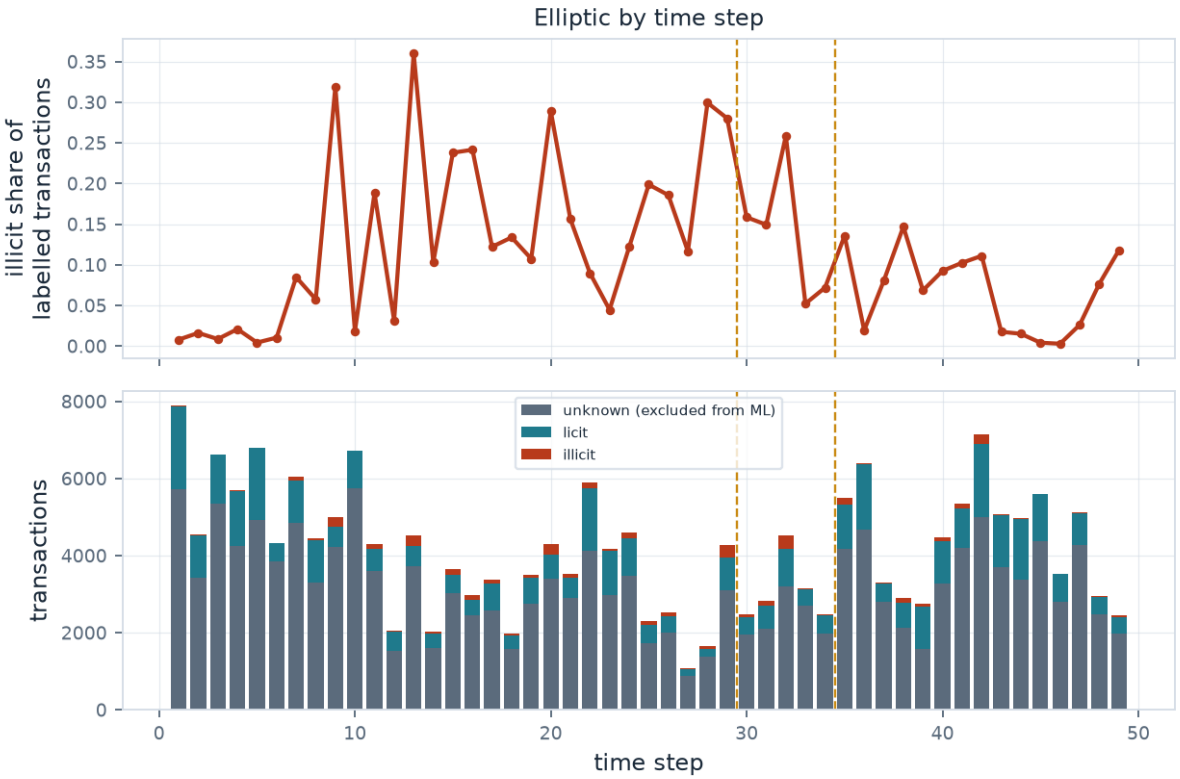
The positive class is an inactivity-based proxy derived from SolRPDS, not an independently verified rug-pull ground truth.



The positive class is an inactivity-based proxy derived from SolRPDS, not an independently verified rug-pull ground truth.







Dashed lines mark the train / validation / test boundaries. Unknown transactions are shown for description only.

7. Label Definitions, Limitations, and Threats to Validity

SolRPDS: proxy-label limitation

SolRPDS provides no independently verified rug-pull label. The target used here is y = (INACTIVITY_STATUS == 'Inactive'), called the inactivity-based rug-pull proxy label. The positive class is an inactivity-based proxy derived from SolRPDS, not an independently verified rug-pull ground truth. A pool can be inactive for benign

reasons (abandoned project, migration, low interest) and a fraudulent pool can still be recorded as active. Because the inactivity status is probably derived from activity timestamps, LAST_SWAP_TIMESTAMP, LAST_SWAP_TX_ID and every feature derived from them are excluded from the primary model. The ablation model adds timestamp-derived duration features and is reported separately because those features may define or strongly determine the outcome.

Real-CATS: report-only address bias

Many criminal Bitcoin addresses have zero observed on-chain transactions and therefore no timestamps, whereas very few benign addresses do. Addresses with zero observed on-chain transactions are marked report-only / no observed on-chain activity. In the Full BTC model a classifier can learn that 'no observed activity = criminal', which is a property of how the dataset was assembled and not a behavioural fraud signal. The activity-filtered BTC model is the headline BTC result. Undated zero-activity addresses cannot be placed on a chronological split. They are kept only in the Full BTC model, as training-only rows, and are scored separately as report-only diagnostics. They are excluded from the activity-filtered BTC model and from the ETH model. Full and activity-filtered metrics are reported separately and never averaged.

Real-CATS: whole-history feature aggregation

Real-CATS features summarise each address's entire observed history (totals, variances, lifetime, activity counts). A split by first_time orders addresses by when they first appeared, but a feature vector can still include behaviour that happened after the split date. Chronological splitting therefore reduces, but does not eliminate, temporal leakage. Class mix may also differ across periods, so each split's class counts are reported.

Elliptic: temporal distribution shift

Transaction behaviour and illicit activity patterns change over time, so a model trained on time steps 1-29 and tuned on 30-34 may degrade on steps 35-49. This is why validation and test metrics are both reported together with their gap. The unknown class (unlabelled transactions) is not treated as licit and is excluded from training, validation and testing; it is used only for descriptive graph and trend statistics.

Why the models are separate

The four datasets describe different entities (Solana pools, Bitcoin addresses, Ethereum addresses, Bitcoin transactions), use different feature spaces, label definitions and time ranges, and are not merged. Each dataset has its own model. Datasets are never merged, scores from different models are not comparable, and no cross-dataset generalisation is claimed.

Why scores are not comparable across datasets

A model risk score is a function of one model, one feature space and one class balance. Class prevalence in each dataset reflects how it was collected, not real-world base rates, and the scores are uncalibrated. A score of 0.9 from one model therefore means nothing relative to a 0.9 from another.

Why SHAP is associative, not causal

SHAP values describe how the trained model's output depends on its inputs. They are associative explanations of the model, not causal claims about why an address, transaction or pool is risky. Correlated features share credit, and features that are artefacts of dataset construction can appear important. Importance rankings should be read together with the leakage controls described above.

General threats to validity

Labels come from dataset curators and carry their own errors. Hyper-parameters are fixed defaults and were not tuned on the test split. The validation split is used for early stopping and threshold selection, so validation metrics are optimistic relative to test metrics. Results describe these datasets and split rules only.

8. Reproducibility

Item	Value
python	3.14.6
platform	Windows-11-10.0.26300-SP0
pandas	3.0.6
numpy	2.5.4
scikit_learn	1.9.1
xgboost	3.4.1
shap	0.53.0
Training summary generated (UTC)	2026-10-15T04:27:10+00:00

Appendix: model features

SolRPDS - Primary model (17 features)

total_added_liquidity, total_removed_liquidity, num_liquidity_adds, num_liquidity_removes, add_to_remove_ratio, log_added_liquidity, log_removed_liquidity, log_add_to_remove_ratio, net_liquidity_signed_log, removed_share_of_flow, count_ratio_adds_to_removes, total_events, log_avg_add_size, log_avg_remove_size, removed_exceeds_added, mint_is_wsol, mint_is_usdc

SolRPDS - Ablation model (21 features)

total_added_liquidity, total_removed_liquidity, num_liquidity_adds, num_liquidity_removes, add_to_remove_ratio, log_added_liquidity, log_removed_liquidity, log_add_to_remove_ratio, net_liquidity_signed_log, removed_share_of_flow, count_ratio_adds_to_removes, total_events, log_avg_add_size, log_avg_remove_size, removed_exceeds_added, mint_is_wsol, mint_is_usdc, pool_active_duration_hours, swap_after_first_pool_hours, swap_after_last_pool_hours, has_swap_timestamp

Real-CATS BTC - Activity-filtered (26 features)

balance, total_received_BTC, total_sent_BTC, total_received_USD, total_sent_USD, transaction_fee, transaction_fee_Variance, transaction_number, payment_transactions, receipt_transactions, lifetime, total_output_slots, total_input_slots, activity_w, activity_d, activity_time, max_sent_amount, min_sent_amount, max_received_amount, min_received_amount, received_Variance_BTC, sent_Variance_BTC, received_Variance_USD, sent_Variance_USD, sent_counters, received_counters

Real-CATS BTC - Full (26 features)

balance, total_received_BTC, total_sent_BTC, total_received_USD, total_sent_USD, transaction_fee, transaction_fee_Variance, transaction_number, payment_transactions, receipt_transactions, lifetime, total_output_slots, total_input_slots, activity_w, activity_d, activity_time, max_sent_amount, min_sent_amount, max_received_amount, min_received_amount, received_Variance_BTC, sent_Variance_BTC, received_Variance_USD, sent_Variance_USD, sent_counters, received_counters

Real-CATS ETH (46 features)

46 features: balance ... to_EOA_wod_normal (local_01-local_93, agg_01-agg_72)

Elliptic (165 features)

165 features: local_01 ... agg_72 (local_01-local_93, agg_01-agg_72)

Elliptic - Ablation (167 features)

167 features: local_01 ... out_degree (local_01-local_93, agg_01-agg_72), in_degree, out_degree